

PROJECT DESIGN PHASE II – DATA FLOW DIAGRAM & USER STORIES

DATE	04/10/2026
TEAM ID	SWTID-2026-8904
PROJECT NAME	AI Study Buddy

MAXIMUM MARKS: 4 Marks

Data Flow Diagram

The DFD shows how student data moves between the user interface, backend, database and AI service.

[STUDENT]	→	[WEB / STUDY DASHBOARD]	→	[NODE.JS + EXPRESS API]
			■	■
		[MONGODB]		[GOOGLE GEMINI AI]

User Stories

Functional Requirement	US	User Story / Task	Acceptance Criteria	Priority
Registration	US-1	As a student, I can register with name, email and password.	User details are saved in MongoDB.	High
Login	US-2	As a student, I can log in securely.	Valid credentials allow login.	High
Dashboard	US-3	As a student, I can access my study dashboard after login.	Dashboard loads after authentication.	High
AI Explanation	US-4	As a student, I can ask a question and receive an AI explanation.	Gemini generates and displays an answer.	High
Quiz	US-5	As a student, I can generate and attempt a quiz.	Quiz questions and score are displayed.	High
Notes	US-6	As a student, I can create and search study notes.	Notes are stored and retrieved correctly.	Medium
Progress	US-7	As a student, I can view study history and progress.	Progress data is displayed from MongoDB.	Medium